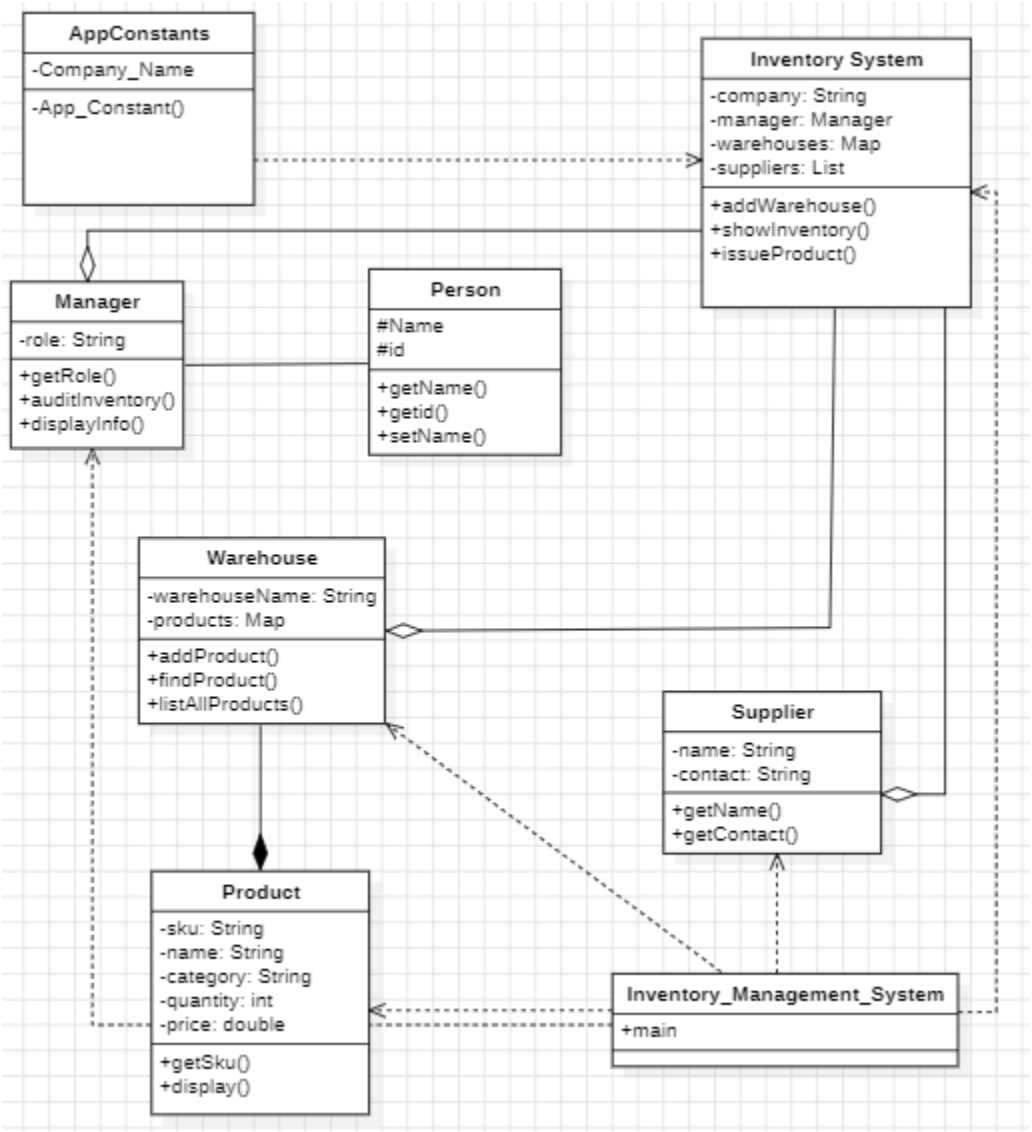
UML Diagram:


AppConstants
-Company_Name
-App_Constant()
Inventory System
-company: String
-manager: Manager
-warehouses: Map
-suppliers: List
+addWarehouse()
+showInventory()
+issueProduct()
Manager
-role: String
+getRole()
+auditInventory()
+displayInfo()
Person
#Name
#id
+getName()
+getid()
+setName()
Warehouse
-warehouseName: String
-products: Map
+addProduct()
+findProduct()
+listAllProducts()
Supplier
-name: String
-contact: String
+getName()
+getContact()
Product
-sku: String
-name: String
-category: String
-quantity: int
-price: double
+getSku()
+display()
Inventory_Management_System
+main

Inventory Management System
UML Diagram Explanation

1. System Overview
The Inventory Management System manages warehouses, products, suppliers, and managerial operations. It follows an object-oriented design with separate classes for each entity to ensure modularity and reusability.

Main Controller	InventorySystem
Managed By	Manager (inherits from Person)
Stores	Warehouses containing Products
Supplied By	Suppliers
Constants	AppConstants (e.g., Company Name)
Entry Point	Inventory_Management_System (main method)

2. Class Explanations
• AppConstants
Stores global constants such as the company name, used across the application.

• Person
Base class with attributes name and id. It provides getters and setters for personal details.

• Manager
Inherits from Person. Oversees inventory, audits stock, and manages warehouses and suppliers.

• InventorySystem
Central class managing all warehouses and suppliers. It handles adding, showing, and issuing products.

• Warehouse
Represents a storage unit containing multiple products in a map structure.

• Product
Represents an individual product with SKU, name, category, quantity, and price.

• Supplier
Represents companies providing products. Stores name and contact information.

• Inventory_Management_System
Main class that initializes and runs the system.

3. Relationships Between Classes

Type	Classes Involved	Explanation
Inheritance	Manager → Person	Manager is a specialized Person.
Composition	Warehouse → Product	Products belong to a specific Warehouse.
Aggregation	InventorySystem → Warehouse, Supplier	System contains multiple warehouses and supp
Association	InventorySystem ↔ Manager	Manager manages the InventorySystem.
Dependency	Inventory_Management_System → All	Main program depends on all other components

4. Example System Workflow

- The system starts using the main() method.
- AppConstants loads the company name.
- A Manager object is created and assigned to the InventorySystem.
- Warehouses are added under the InventorySystem.
- Products are added into specific warehouses.
- Suppliers are registered with contact details.
- The Manager audits inventory and can issue products if needed.
- System displays updated stock and supplier details.

5. Summary

The UML diagram models a structured and maintainable inventory system where each class has a clear role. Using inheritance, composition, and aggregation ensures modularity and reusability. The Manager oversees the system, while Warehouses, Products, and Suppliers interact through the InventorySystem core.